C++ Home Tasks - Solutions

Task 1: Swap two variables

Write a program that reads two variables and swaps their values.

Step 1: Declare three integers: var1, var2 and a helper variable `temp`.
Step 2: Read var1 and var2 from the user using cin.
Step 3: Copy var1 into temp: `temp = var1;` (so the old value is not lost).
Step 4: Copy var2 into var1: `var1 = var2;`
Step 5: Copy temp into var2: `var2 = temp;` Now the values are swapped.
Step 6: Print both variables after the swap.

Code

```
#include <iostream>
using namespace std;

int main()
{
    int var1, var2, temp;

    cout << "Enter value of var1: ";
    cin >> var1;
    cout << "Enter value of var2: ";
    cin >> var2;

    cout << "Before swap: var1 = " << var1 << ", var2 = " << var2 << endl;

    temp = var1;
    var1 = var2;
    var2 = temp;

    cout << "After swap: var1 = " << var1 << ", var2 = " << var2 << endl;
    return 0;
}
```

Output (input: 10 and 20)

```
Enter value of var1: 10
Enter value of var2: 20
Before swap: var1 = 10, var2 = 20
After swap: var1 = 20, var2 = 10
```

Task 2: Small letter to capital letter

Read an alphabet in small letters (a-z) and display it in capital letters (A-Z). Hint: ASCII.

Step 1: Declare two char variables: small and capital.
Step 2: Read the small letter with cin.
Step 3: In ASCII, 'a' = 97 and 'A' = 65, so the difference is 32.
Step 4: Subtract 32 from the small letter: `capital = small - 32;`
Step 5: Print the capital letter.

Code

```
#include <iostream>
using namespace std;

int main()
{
    char small, capital;

    cout << "Enter an alphabet in small letters (a-z): ";
    cin >> small;

    capital = small - 32;   // ASCII: 'a' = 97, 'A' = 65, difference = 32

    cout << "Upper case: " << capital << endl;
    return 0;
}
```

Output (input: g)

```
Enter an alphabet in small letters (a-z): g
Upper case: G
```